

VOCAL EXPRESSION ASSIGNMENT FOR STRUCTURAL PRIMITIVES

May 7, 2026

0.1 Inscribing Grammar — Phonetic Inscription

Each primitive is assigned a vocal articulation whose physics mirrors the structural meaning of the primitive. The IPA symbol(s) in brackets give the primary vocal expression.

0.2 D (Dimensionality)

- **D_wedge**: glottal stop [ʔ] — point of complete closure, 0d articulatory event
- **D_triangle**: bilabial fricative [ɸ] — surface contact of two planes
- **D_infty**: vowel continuum [au] — infinite-dimensional formant space
- **D_odot**: click [ǀ] — self-referential implosive release, inward-then-outward

0.3 T (Topology)

- **T_network**: trilled rhotic [r̥] — branching resonances across articulation points
- **T_in**: ingressive breath-hum [h̥] — inclusion, air drawn inward through constriction
- **T_bowtie**: affricate [tʃ] — crossing: stop→fricative transition point
- **T_boxtimes**: compound consonant cluster [kp] — irreducible co-production
- **T_odot**: ejective loop [kʰ] — self-closing articulatory circuit

0.4 R (Relational mode)

- **R_super**: supervisory pitch rise [↗] — one direction: higher structure supervenes on lower
- **R_cat**: isometric vowel pair [↔] — functorial mapping between positions
- **R_dagger**: voiced-voiceless pair [bp] — adjoint pairing within one place
- **R_lr**: bidirectional sigh [h↔h] — two-way acoustic coupling

0.5 P (Parity/Symmetry)

- **P_asym**: asymmetric rasp [ɹ] — no symmetry preserved
- **P_psi**: biphonic overtone [+̎] — superposed fundamental+overtone
- **P_pm**: voicing pair [z↔s] — Z2 voiced/voiceless flip
- **P_sym**: neutral schwa [ə] — all symmetries preserved, maximal neutrality
- **P_pm_sym**: Frobenius snap [ts] — $\mu\delta=\text{id}$ at the critical crossing point

0.6 F (Fidelity)

- **F_ell**: deterministic tap [] — classical, no coherence needed
- **F_eth**: breathy voice [] — thermal noise mixed with phonation
- **F_hbar**: coherent whistle [ç] — quantum-phase-stable laminar tone

0.7 K (Kinetics)

- **K_fast**: driven plosive [k] — rapid energy release
- **K_mod**: moderate approximant [] — intermediate timescale
- **K_slow**: prolonged nasal hum [ŋ] — near-equilibrium sustained tone
- **K_trap**: frozen-order formant [a] — trapped ordered resonance
- **K_MBL**: creaky voice [a] — frozen disorder, irregular vibration

0.8 G (Scope)

- **G_beth**: local dental tap [] — nearest-neighbor articulation
- **G_gimel**: mesoscale pharyngeal [] — intermediate range constriction
- **G_aleph**: universal glottal [h] — long-range, whole-tract participation

0.9 Gamma (Interaction grammar)

- **G_and**: simultaneous labial-velar [kp] — conjunctive co-articulation
- **G_or**: fluctuating rhotic [r] — alternate articulator paths
- **G_seq**: sequential diphthong [] — ordered temporal progression
- **G_broad**: broadcast exclamation [a] — one-to-all projection

0.10 Phi (Criticality)

- **Phi_sub**: dead stop [] — no criticality, no scaling
- **Phi_c**: critical creak [] — self-modeling gate, critical divergence
- **Phi_c_complex**: vibrato [a] — complex-plane oscillation
- **Phi_EP**: exceptional merge [a] — non-Hermitian degeneracy, two modes coalesce
- **Phi_super**: runaway howl [a] — supercritical, no bounded attractor

0.11 H (Temporal depth)

- **H0**: memoryless click [] — no prior state needed
- **H1**: one-step echo [a.] — one temporal memory
- **H2**: two-step cadence [a..] — two prior states retained
- **H_inf**: eternal drone [o] — no finite Markov boundary

0.12 S (Stoichiometry)

- **one_one**: unique phoneme [] — one type, one instance
- **n_n**: reduplicated mora [ma.ma] — many identical units
- **n_m**: heterogeneous syllable [st] — multiple distinct phoneme types

0.13 Omega (Winding)

- **Omega_0**: unwound utterance [pa] — trivial topology
- **Omega_Z2**: binary parity chirp [ppaa] — Z2 protected, requires H2+

- **Omega_Z**: winding hum $[\tilde{a} \searrow \tilde{a} \nearrow \tilde{a}]$ — integer topological winding number
- **Omega_NA**: non-Abelian braid $[\leftrightarrow]$ — path-dependent braiding requires $D_{\odot}$